

Sol Data Flow Diagrams

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❓ Sol — Sơ đồ luồng dữ liệu theo thời gian + chức năng

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Render: Mở file này trong browser — Mermaid auto-render diagrams.

Mục lục:

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- 4. Flow: Daily push cron fire
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1. Timeline Data Flow — 52 ngày

Sơ đồ Gantt-style cho 52 ngày journey. Mỗi swimlane = 1 loại data flow.

❓ **USER INPUT** — User gửi vào hệ thống

❓ **SOL OUTPUT** — Sol gửi cho user

🔗 **DB WRITE** — Hệ thống ghi vào DB

🔗 **CRISIS** — Crisis event

ganttt title 52-Day Journey Data Flow Timeline dateFormat YYYY-MM-DD axisFormat %d/%m section Phase 1 (D0) Welcome message (follow event) :crit, p1a, 2026-04-29, 1h User click button "Tôi vừa quyết tâm" :p1b, after p1a, 2h User pick Q-Day :p1c, after p1b, 5h Enroll → 52 ScheduledPush created :crit, p1d, after p1c, 6h section Phase 2A LamQuen T-21 Daily chip "Vi sao bạn nghiện" :p2a1, 2026-04-30, 1d T-18 FTND quiz :p2a2, 2026-05-03, 1d T-15 Cam kết bridge :crit, p2a3, 2026-05-06, 1d section Phase 2B GiamDan T-14 Giảm 25% daily :p2b1, 2026-05-07, 1d T-10 Cắt điều sáng :crit, p2b2, 2026-05-11, 1d T-7 Mốc 1 tuần :p2b3, 2026-05-14, 1d T-3 Vứt thuốc :crit, p2b4, 2026-05-18, 1d T-1 Đêm trước Q-Day :crit, p2b5, 2026-05-20, 1d section Phase 3 Q-Day D0 Q-Day morning :crit, p3a, 2026-05-21, 12h SOS spike + crisis events :crit, p3b, 2026-05-21, 12h section Phase 4 Q-Day Series D3 Bức Tường (peak crisis) :crit, p4a, 2026-05-24, 1d D7 Mốc 1 tuần :p4b, 2026-05-28, 1d D14 Mốc 2 tuần :p4c, 2026-06-04, 1d D16 Đi nhậu lần đầu :crit, p4d, 2026-06-06, 1d D21 Mốc 3 tuần :p4e, 2026-06-11, 1d D24 Identity shift :p4f, 2026-06-14, 1d D30 Graduation :crit, p4g, 2026-06-20, 1d section Phase 5 Weekly maintenance (D31+) :p5, 2026-06-21, 30d D60 Anniversary :p5b, 2026-07-20, 1d D90 X-ray reminder :p5c, 2026-08-19, 1d

2. Hướng luồng dữ liệu (in/out/internal)

Sơ đồ tổng quan 3 hướng dữ liệu giữa User, Sol Backend, và External services.

```
flowchart LR
    subgraph User [USER]
        Z[Zalo OA app]
        U[Web sol.vn]
        A[App bothuocla.sol.vn]
    end
    subgraph SolBackend [SOL BACKEND]
        direction TB
        WH[Webhook]
        API[REST API]
        U1[api/users, api/checkins...]
        CRON[Cron Worker]
        DB[(PostgreSQL 24 jobs)]
    end
    WH --> DB
    DB --> CRON
    CRON --> DB
    subgraph External [EXTERNAL]
        ZL[Zalo OpenAPI]
        ZNS[ZNS + Free OA]
        EM[Email SMTP]
        TG[Telegram Bot]
    end
    U -- "follow, send_text click_button" --> WH
    WH -- "POST checkin, POST lapse" --> API
    API -- "Click wiki link" --> DB
    DB -- "CRON CRON" --> ZNS
    ZNS -- "ZNS send" --> ZL
    ZL -- "Daily chip, milestone, SOS" --> U
    U -- "Email magic link" --> EM
    EM -- "EM EM" --> DB
    DB -- "CRON CRON" --> SOS[CRITICAL ALERT]
    SOS -- "Push notification" --> User
    style User fill:#dbefe8,stroke:#2563eb
    style SolBackend fill:#fef3c7,stroke:#b85c1c
    style External fill:#dcf7e7,stroke:#16a34a
    style DB fill:#f3e8ff,stroke:#9333ea
```

3. Flow: User follow OA → Enroll (Day 0)

Sequence diagram chi tiết — từ lúc user follow đến lúc 52 push được tạo.

```
sequenceDiagram
    actor U as User participant Z as Zalo OA participant W as Webhook participant J as journeyEngine participant D as Database participant O as oaClient
    U->>Z: Click "Quan tâm OA Đi Cùng Sol"
    Z->>W: POST /webhook { event: "follow", sender_id }
    W->>D: UPSERT ZaloOAUser (zaloUserId, oald, displayName)
    W->>O: oaSendText(welcome + 3 buttons)
    O->>Z: Send Free OA message
    Z->>U: Hiện tin chào + 3 button
    U->>Z: Click "Tôi vừa quyết tâm"
    Z->>W: POST /webhook { event: "user_send_text", text: "/lo-trinh-full-51" }
    W->>D: CREATE User (anonymous-first)
    W->>D: UPDATE ZaloOAUser.userId (LINK)
    W->>D: UPSERT UserState (flow=journey_choice, step=waiting_qday)
    W->>O: oaSendText("Hỏi Q-Day" + 3 button)
    O->>Z: Send Free OA
    Z->>U: Hiện tin hỏi Q-Day
    U->>Z: Click "Ngày mai"
    Z->>W: POST /webhook { text: "/q-day-tomorrow" }
    W->>D: READ UserState (lấy journeyType)
    W->>J: enrollUser({ userId, journeyType, qDayDate })
    J->>D: UPDATE User (journeyType, qDayDate, journeyStatus='active', journeyEnrolledAt)
    loop For each dayOffset (-21 to 30)
        J->>J: resolveTemplateForDay(dayOffset, journeyType)
        J->>J: computeScheduledAt(qDay + offset + preferredHour)
        J->>D: UPSERT ScheduledPush (userId, dayOffset, templateCode, scheduledAt, params)
    end
    J->>W: { created: 52 }
    W->>D: UPDATE UserState (clear stateData)
    W->>O: oaSendText("Đã set Q-Day. 52 lịch tin tạo")
    O->>Z: Send Free OA
    Z->>U: Hiện tin confirm Note over D: DB ghi ~ 56 rows
    (1 ZaloOAUser, 1 User, 1 UserState, 52 ScheduledPush, + ZNSLog)
```

4. Flow: Daily push cron fire (mỗi 5 phút)

sequenceDiagram participant C as Cron */5 min participant F as fireDuePushes participant D as Database participant T as ZaloTemplate participant O as oaClient participant Z as Zalo ZNS API actor U as ? User C->>F: Trigger F->>D: SELECT ScheduledPush WHERE scheduledAt <= NOW() AND status='pending' LIMIT 50 D-->>F: 17 pending pushes loop For each push F->>D: READ User + messagingProfile (mute check) alt User muted/paused F->>D: UPDATE ScheduledPush.status='cancelled' else User active F->>D: UPDATE ScheduledPush.status='sending' (lock) F->>T: READ ZaloTemplate WHERE code=? alt Template chưa APPROVED F->>D: UPDATE status='failed' errorCode='TEMPLATE_NOT_APPROVED' else Template OK F->>O: znsSendTemplate(phone, templateId, params) O->>Z: POST /message/template Z-->>O: { ok, messageId } alt Success F->>D: CREATE ZNSLog (status=SENT) F->>D: UPDATE ScheduledPush.status='sent', znsLogId, sentAt F->>D: UPSERT UserMessagingProfile.totalSent++ Z-->>U: User nhận tin daily chip else Fail F->>D: UPDATE ScheduledPush.status='pending' retryCount++ scheduledAt += backoff Note over F: Exponential backoff: 2, 4, 8 phút end end end end F-->>C: { scanned, sent, failed, skipped }

? **Note:** Cron mỗi 5 phút mới fire. Nếu user chọn preferredPushHour=7, push sẽ thực sự fire trong khoảng 7:00-7:05.

4b. Flow: Web Widget Chat — Anonymous-first auth ?

Khác với Zalo OA (cần follow), Web Widget cho user vào ngay không cần phone/OTP.

sequenceDiagram actor U as ? User participant B as ? Browser (sol.vn) participant W as Sol Widget (IIFE embed) participant LS as localStorage participant API as Sol Backend participant DB as Database participant S as Socket.IO U->>B: Vào sol.vn/cai-thuoc-la-tai-nha B->>W: window.SOLWidget.init() W->>LS: Read sol_token + deviceId alt Token tồn tại (returning user) W->>API: GET /users/me (Authorization: Bearer) API-->>W: User profile else No token (first visit) W->>LS: Generate deviceId (UUID v4) W->>API: POST /auth/anonymous { deviceId, originDomain } API->>DB: CREATE User (isAnonymous=true) API-->>W: { jwt, user } W->>LS: setItem('sol_token', jwt) else URL ?zalo=success&token=XXX (OAuth callback) W->>LS: setItem('sol_token', token from URL) W->>API: GET /users/me W->>B: history.replaceState (clear URL) end W->>S: io(socketBase, { auth: { token } }) S->>API: Verify JWT, join room user:{id} Note over W,U: Widget ready — bubble góc dưới U->>W: Click bubble → mở Panel U->>W: Chat "anh sắp hút" W->>API: POST /messages { text: "anh sắp hút" } API->>DB: WRITE Message (role=USER) API->>API: intentRouter → CRISIS detect API->>S: emit message:new (Sol stream chunks) S->>W: Typing animation realtime W->>U: "Hít sâu 4-7-8. Cơn thêm 90 giây..." API->>DB: WRITE Message (role=ASSISTANT) API->>DB: WRITE CrisisEvent Note over W: Sau 3 ngày engagement W->>U: Soft prompt "Liên kết Zalo bảo vệ hành trình"

? **Key insight:** Widget = **zero-friction entry**. User chỉ cần MỞ TRANG là vào được. Khác Zalo OA phải follow + cho phép push.

4c. Flow: State sync 3 surface ?

User có thể access Sol qua 3 nơi cùng lúc. Sol đồng bộ data realtime.

flowchart TB subgraph Surfaces["3 Surface — Same User"] W1["? Widget PC (sol.vn wiki page)"] W2["? Dashboard mobile (bothuocla.sol.vn)"] W3["? Zalo OA chat (Zalo app)"] end subgraph Backend["Sol Backend"] API["REST API + Socket.IO"] DB["User cmp1 messages, checkins, journey state"] end W1 -- "deviceId_A" --> API W2 -- "deviceId_B (or zaloUserId)" --> API W3 -- "zaloUserId_X" --> API API <--> DB API -- "emit message:new checkin:reminder" --> W1 API -- "emit message:new checkin:reminder" --> W2 API -- "ZNS push" --> W3 style W1 fill:#d9eaf7,stroke:#25538b style W2 fill:#d9eaf7,stroke:#25538b style W3 fill:#f3f3f3,stroke:#b85c1c style DB fill:#f3e8ff,stroke:#9333ea

Identity binding flow – Merge accounts

sequenceDiagram actor U as ? User participant W as Widget PC participant Z as Zalo Mobile participant API as Sol Backend participant DB as Database Note over U: Lần 1: PC U-»W: Vào sol.vn → anonymous W-»API: POST /auth/anonymous { deviceId_A } API-»DB: CREATE User cmp1 (anon, deviceId=A) Note over U: Lần 2: Mobile U-»Z: Follow OA Sol trên Zalo Z-»API: Webhook follow event API-»DB: CREATE ZaloOAUser (zaloUserId=X) API-»DB: (Optional) CREATE User cmp2 (anon, zaloUserId=X) Note over U: Bind Zalo trên Widget PC U-»W: Click "Liên kết Zalo" W-»API: Redirect Zalo OAuth API-»W: Callback ?token=XXX (cmp2's JWT) Note over API: MERGE LOGIC API-»DB: READ User cmp1 (current widget session) API-»DB: READ User cmp2 (zaloUserId=X) API-»DB: MERGE: cmp1.zaloUserId = X API-»DB: UPDATE ZaloOAUser.userId = cmp1 API-»DB: COPY messages, checkins từ cmp2 → cmp1 API-»DB: DELETE cmp2 Note over U: Lần 3: PC khác (login Zalo) U-»W: PC mới login bằng Zalo W-»API: OAuth → token cmp1's API-»W: User cmp1 full state W-»U: Restore tất cả hành trình

? **Khi merge xảy ra:** Backend đảm bảo messages, checkins, journey state được copy sang account chính. User KHÔNG mất data dù dùng 5 thiết bị khác nhau.

Touchpoint Widget Chat – 15 events chính

Touchpoint	UI	API endpoint	Lưu vào
Widget mount (lần đầu)	Bubble góc	POST /auth/anonymous	User (anon)
User chat text	Input box	POST /messages	Message (USER)
Sol AI stream reply	Typing indicator	SSE /messages/stream	Message (ASSISTANT)
Daily check-in	Flow 4 step	POST /checkins	CheckIn
Click chip suggestion	Pre-fill prompt	POST /messages	Message + CannedReply++
90s breathing timer	Modal overlay	POST /crisis-timer	CrisisTimerLog
Log "Tôi đã hút"	Lapse modal	POST /lapse	LapseEvent + CigaretteLog
Workbook entry	Journal editor	POST /journal	ProgressJournal
Listen Khang voice	Audio inline	POST /voice/listen	KhangVoiceListen
FTND test 6 câu	Quiz UI	PUT /users/me/profile	User.ftndScore
Quit reasons (3-5)	Wizard	PUT /users/me/profile	User.quitReasons[]
Top triggers	Wizard checkbox	PUT /users/me/profile	User.topTriggers[]
Bind Zalo	OAuth button	POST /auth/zalo/callback	User.zaloUserId + MERGE
Bind phone OTP	Modal nhập SĐT	POST /auth/phone/{request,verify}	User.phone + OtpCode
Unbind / Logout	Settings	POST /auth/logout	Clear localStorage

Widget vs Zalo OA – So sánh chiến lược

Tiêu chí	? Web Widget	? Zalo OA
Auth barrier	Anonymous-first (zero friction)	Phải follow OA
Push proactive	Web Push + Socket realtime	ZNS template (mạnh hơn về reach)
Reply latency	< 200ms (SSE stream)	< 5s (Zalo API)
Cost mỗi tin	0đ	~ 300đ/tin ZNS
Phone notification	Chỉ khi browser mở	? Push mạnh
Stream typing	? AI thật chunk	? Fixed template
Rich UI	? React full	Text + 3 button max
Voice playback	? Inline player	Phải click URL ra browser
Workbook journal	? Editor đầy đủ	Chỉ chat text
Crisis 90s timer	? Visual countdown	Text "đếm 1,2,3..."

? Best practice:

- **Zalo OA** → cho **push** (Sol nhắc anh hằng ngày qua phone notification)
- **Web Widget** → cho **deep work** (workbook, voice, AI chat sâu, checkin chi tiết)

User dùng cả 2 cùng lúc – Zalo nhận push, click link wiki → vào sol.vn → widget tự nhận diện user (qua bind Zalo) → tiếp tục chat sâu hơn.

5. Flow: SOS Trigger + Response

sequenceDiagram actor U as ? User participant Z as Zalo OA participant W as Webhook participant I as intentRouter participant S as sosHandler participant D as Database participant O as oaClient participant SK as Socket Admin participant K as ? Khang Admin alt SOS via button click U->>Z: Click [?] Tôi đang khó Z->>W: text="/sos" W->>S: triggerSos(triggerType=button) else SOS via keyword U->>Z: Chat "Tôi sắp hút lại rồi" Z->>W: text="..." W->>I: routeUserMessage(text) I->>I: Match CRISIS keyword I->>W: { type: 'CRISIS', matchedKeyword: 'sắp hút' } W->>S: triggerSos(triggerType=keyword) end S->>S: classifySeverity(keyword/message) Note over S: critical | high | medium | low S->>D: READ User (lookup name, currentDay) S->>D: CREATE SOSAlert (severity, triggerType, userMessage) S->>O: oaSendText(SOS reply template) O->>Z: Free OA reply Z->>U: "Đừng làm gì 90s. Hotline 02439931800" S->>D: UPDATE SOSAlert.status='auto_responded' S->>SK: socket.broadcast('sos:new', payload) SK->>K: ? ALERT trên admin dashboard real-time alt Severity=critical (TODO Sprint 4) S-->>K: Telegram alert qua bot end alt User click [Tôi đã vượt qua] U->>Z: text="/victory" Z->>W: Webhook W->>S: handleVictory() S->>D: UPDATE SOSAlert.status='resolved', resolvedAt S->>O: oaSendText("Tuyệt vời! Anh đã vượt cơn") O->>Z: Free OA Z->>U: Tin chúc mừng else Khang reply qua admin K->>SK: POST /sos/:id/respond SK->>D: UPDATE SOSAlert.status='admin_responding' respondedAt, responseMessage SK->>O: oaSendText(Khang's message) O->>Z: Free OA Z->>U: Khang reply else 72h không respond Note over D: Cron daily check UPDATE status='no_response_72h' end

6. Flow: Lapse Event Recovery

Khi user "Tôi đã hút 1 điếu" (lapse) — Sol chuyển sang recovery mode (KHÔNG kết tội, KHÔNG cancel journey).

sequenceDiagram actor U as ? User participant A as App bothuocla.sol.vn participant API as /api/lapse participant D as Database participant J as journeyEngine participant O as oaClient participant Z as Zalo OA U->>A: Click "Tôi đã hút 1 điếu" (lapse log) A->>API: POST /api/lapse { count: 1, time, context } API->>D: CREATE LapseEvent (userId, time, count, context) API->>D: CREATE CigaretteLog (userId, count, day) alt Lapse > threshold (vd 3 điếu/ngày) API->>J: Đánh dấu relapsed J->>D: UPDATE User.journeyStatus='relapsed' J->>D: UPDATE ScheduledPush WHERE status='pending' SET status='cancelled' Note over J: Re-enroll vào shorter recovery journey (TBD) end API->>O: oaSendText(SOL_LAPSE_RECOVERY template) O->>Z: ZNS template Z->>U: "1 lapse ≠ relapse. 70% phục hồi (Marlatt). Tiếp tục anh nhé"

? **Triết lý lapse:** Theo Marlatt 1985, 70% user phục hồi sau 1 lapse nếu KHÔNG bị tự đánh giá. Sol không cancel journey — chỉ note + encouragement.

7. State machine: JourneyStatus (per User)

stateDiagram-v2 [*] --> NULL: User mới NULL --> active: enrollUser() active --> paused: User pause (Cancel pending push) paused --> active: User resume (Re-schedule push) active --> relapsed: Lapse > threshold relapsed --> active: Re-enroll (shorter recovery) active --> graduated: D30 hit graduated --> [*]: Maintenance mode note right of active Cron 7:30 AM: currentJourneyDay updated end note note left of paused ScheduledPush: pending → cancelled end note